

A Packing of 6 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 6 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.879908. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.799800, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 6 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.879908 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 6$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.799800, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 6/5 = 2.880000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N6.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
719977059306476627628145185574044087970862270348185738071237401636206073
811319159317493966071991417007812649174968172946247287445567724435465511
344540381162642894516703943650420323880918750900
```

denominator:

```
249999999999819727612466557509128529103187137513251806121126918233600326
901509464024607892248415900223112085945913177330347389517778273814991637
466206170906192759974057098716072801311559328347
```

B Coordinates

6 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N6.txt`.

```
-288794830237 -640778108848 -16954058927 -2680026207108193036959 266965798538 237530892146 -1404258338681485142710 -44846340035 -173750720179 -151394251759
-176104559586 55370380312 -125649861296 50282365966466402732599 -283299803257 -565986610883 234028248722938023734 -176139115661 215673341699 -351505775878
553231036136 -44050354833 -538382426971 84729516146431071286672 -10690865268 -92416554473 -40816818037420709417793 -6366850090 555989596604 -258444262542
250062718413 -277106672281 570085791930 3303303194940873647516 117710546387 107352121469 -58061296143-91853427125 429295635806 -133084427374 -275470199922
-469409137167 -168771761076 -526920920973 2609952242840200814 -535977449014 220745021433 291698221720480131702117 430095577283 71393304570 84464324731
227514672834 59778890041 -44200618331 370095262762 -522491675077 -266484734745 389295360803 -283801928119730499484 -702213284125 -82139350562 -139201388578
314061428542 -580818678980 -148479346135 2354820083224106876262 350963895897 -246144639256 49782463368786815667480 -260341381784 160348386143 321289604557
-90043631627 -218935952800 233931724862 25415236784842900689 -63415723783 17058362016 -230691965686 306903834197 -63600409236 -517118874456 460025783221
-273086385078 -179414595933 69147722909 34660803809-220910117870 149296892992 -522504271797 -521012057794858403329 86396665495 -441184906102 -447414931656
252115673882 -240995197205 -88996050932 -48240042186627095062357 571601001003 262687659449 13193137523411566369608 -158920750957 -311204542220 -411465795839
-151563758144 154948507198 653299646332 -2765434268239031703966 438974530344 -58666886807 30481462908243802786965 -529543856284 111716201671 291351121703
76223251634 -156414351212 317992166995 349888679610-177493237371 189527598054 465117852180 4503779055426843853609 84173304716 283546833361 -452282624716
-120268718430 -272405301811 185990697003 -120233159269153721559 175713915577 280921982387 3543396699263589349839 131543051013 -384648111548 410004457827
65932314775 -140851521590 491768465148 -29408842478-666983373637 -211392070488 33660782905 491655340714604263050468 262975090347 -118998581683 78327033693
-114392342282 -270202929745 69171897447 -25108242483792491909 -134737625757 -363960664640 142121041740275221531605 -159149875060 685825226954 239695601273
-487356520142 -176047482459 -76586185484 -5566888193964447163717 -129370170231 -172539943656 4020762976201175978177 -383020362073 -326102568146 -314040999320
816727889817 -180679029417 102210948125 -42683775995012221345625 -71038178782 60963096961 -32238733676485807769296 -366301000986 -60343789356 -225460954919
-115738062992 -675230790935 -78475749171 4209498811334352164652 667870742269 135142482344 -3206043139033204152404 367588863806 -27475984123 344532575924
122725563675 43644734960 -395358236672 -39751080651448214079777 -447323765359 652018995706 -14978901836774853089864 -153921228580 249742700129 348416066320
79394668946 -512828408293 -109945165946 266192832322096982881845 -98648328283 -301375535560 2932468129449582937261 401267573339 318651856500 -298850042259
```